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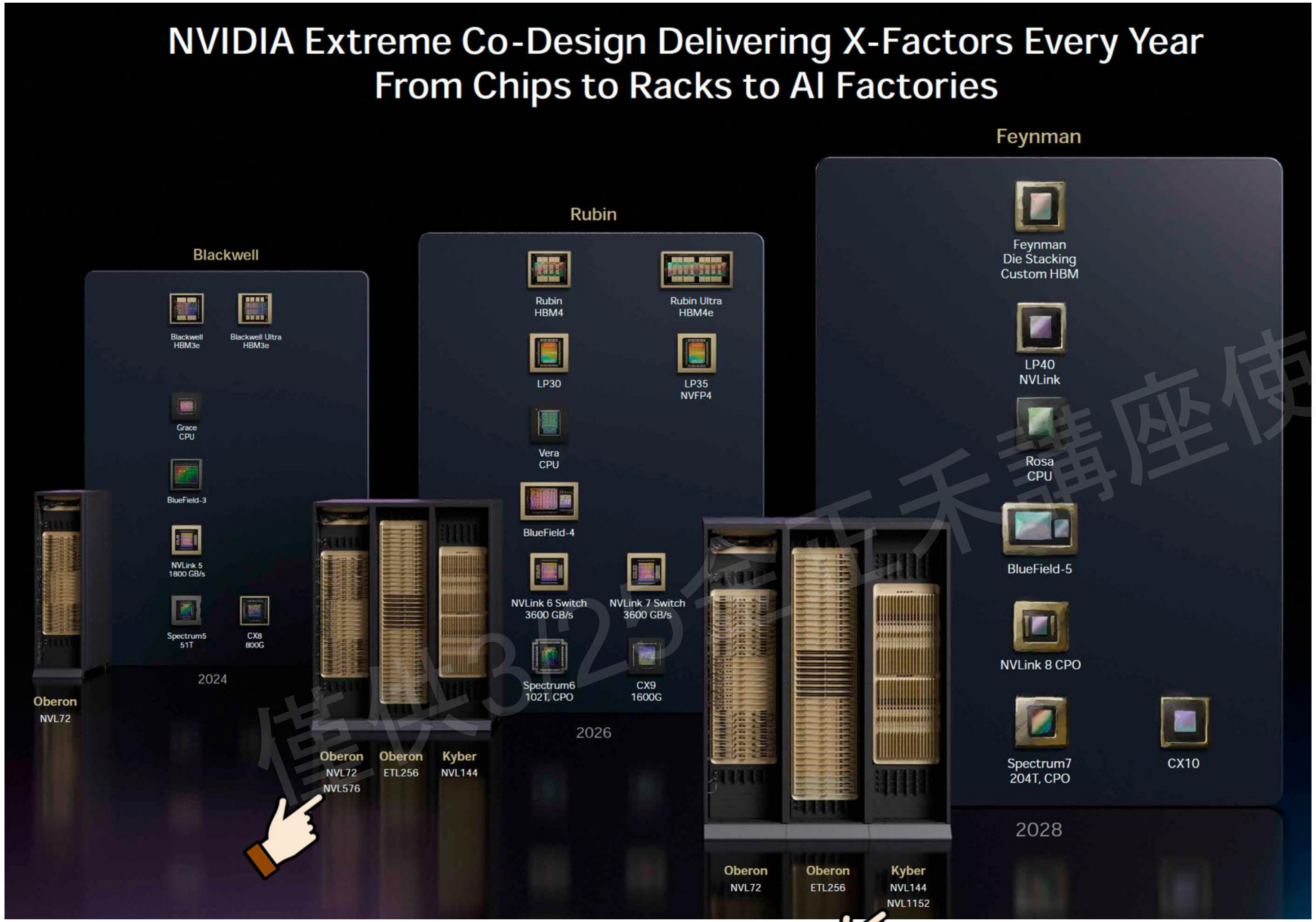
The Scale-Up CPO Era: How Optoelectronic Co-Packaging Breaks Through AI Infrastructure Bottlenecks

[25/03/2026]

NVDIA Roadmap in GTC 2026
CPO is Promised in The Future



VeraRubin Era(2026)

Oberon Structure-

- NVL72: Copper Scale-Up
- NVL576: Optical Scale-Up

Kyber Structure-

- NVL144: Copper Scale-Up

RosaFeyman Era(2028)

Oberon Structure-

- NVL72: Copper Scale-Up

Kyber-

- NVL144: Copper Scale-Up
- NVL1152: CPO based optical Scale-Up

NV will do Both Optical/ Electrical Scale-Up & Optical Scale-Out in the future.

Source: 2026 NVIDIA GTC

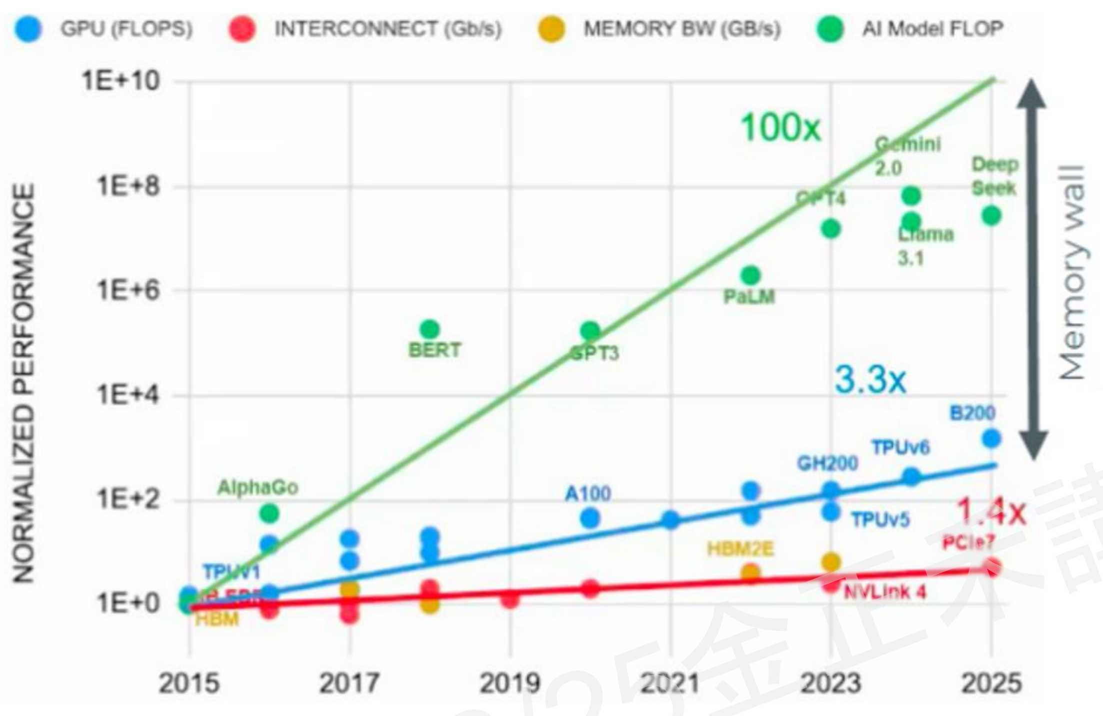
Content

Core Challenges of Scale-Up and the Inevitability of CPO

1

Overcoming the Memory Wall

Unleashing GPU performance requires a massive leap in interconnect scaling.



- AI models grow 100x every 2 years
- Compute performance improves 3.3x every 2 years

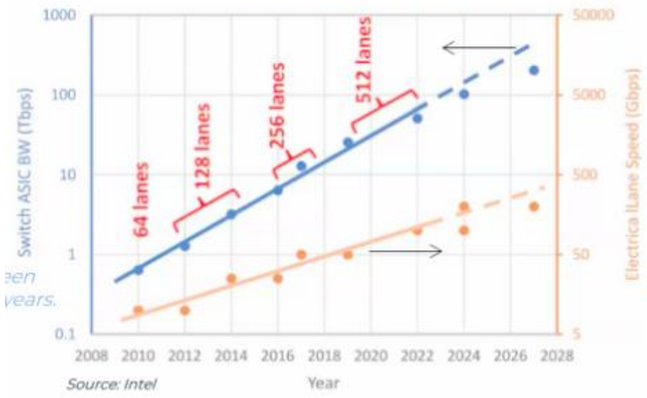
NEED 30x MORE COMPUTE infrastructure

- Compute performance improves 3.3x every 2 years
- Interconnect bandwidth improves 1.4x every 2 years
- Memory bandwidth improves 1.4x every 2 years

NEED 2.4x MORE INTERCONNECTS (both copper and optical)

In total:
NEED 70x MORE INTERCONNECTS deployed every 2 years

Higher RADIX is a useful solution
→ switch bandwidth scaling



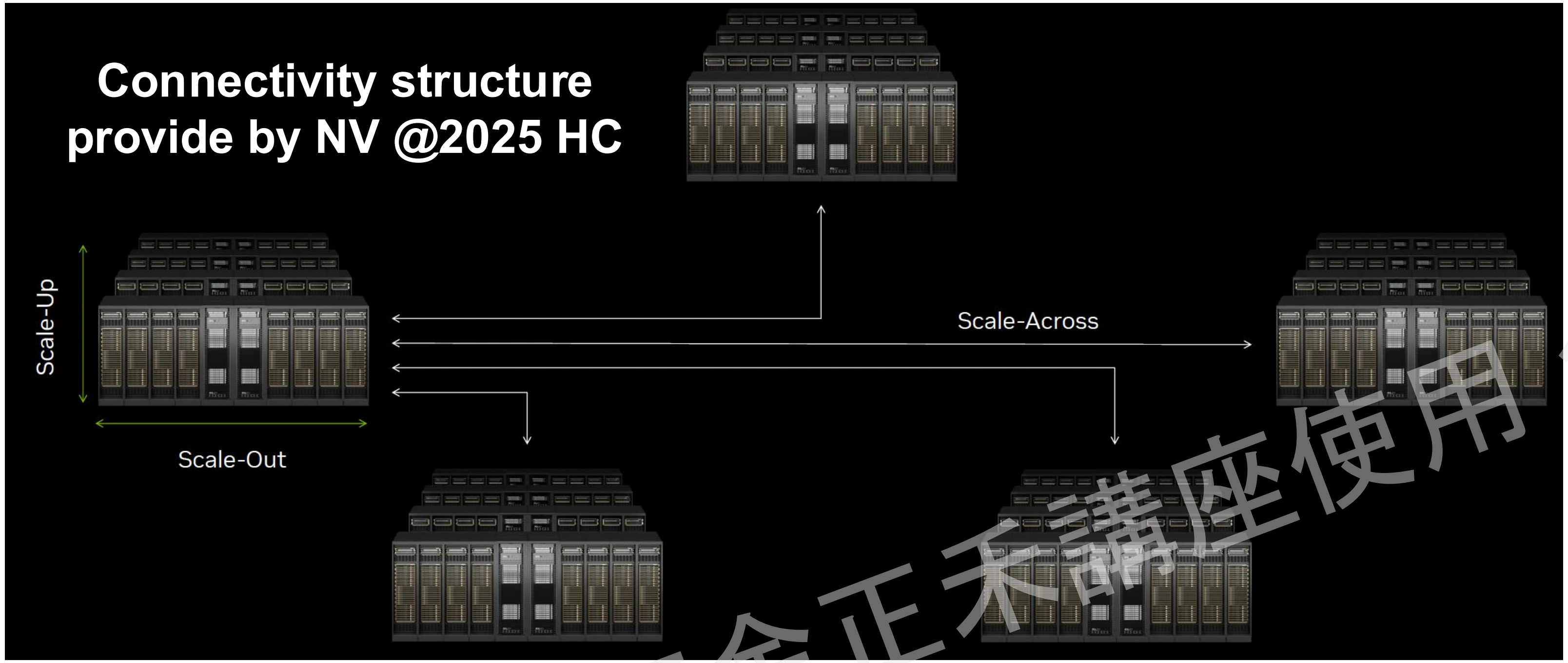
Lane speed has been doubling every 4 years.

<u>Switch bandwidth scaling</u>

- If the lane speed doesn't double, then, to keep up with the required switch performance, the number of lanes must be doubled.
  - Every 2 years, the number of lanes doubles

•**Bandwidth Scaling:** Relies on faster lane speeds (112G→224G→448G) or adding more physical lanes.
•**The Bottleneck:** Strictly limited by Signal Integrity (SI) and chassis physical space.
CPO emerges to break these physical barriers.

Defining the Battlefield: The Hierarchy of AI Connectivity
Some Work Can Only Be Done Within a Single Rack



	Scale-Up (Intra Rack Connection)	Scale-Out (Intra Cluster Connection)	Scale-Across (Inter Cluster Connection)
Reach	2m-5m	500m-2Km	>100Km
Current solution	Electrical(e.g.Nvlink)	Optical(PAM)	Optical(Coherent)
#xPU connected	>72	>500	>10K
Bandwidth/xPU	~7200bit/s	~800bit/s	~100bit/s
Latency	100nS	~uS	~mS

GPUs can work together for specific tasks (Tensor Parallelism)

*Spec. above is limited by current solution

The Ultimate Goal

Massively Connect xPUs
Seamlessly scale workloads across racks, clusters, and data centers as a single logical xPU

Maintain Sufficient Bandwidth
Narrow the bandwidth gap from intra-rack to inter-cluster links to keep xPUs fully utilized.

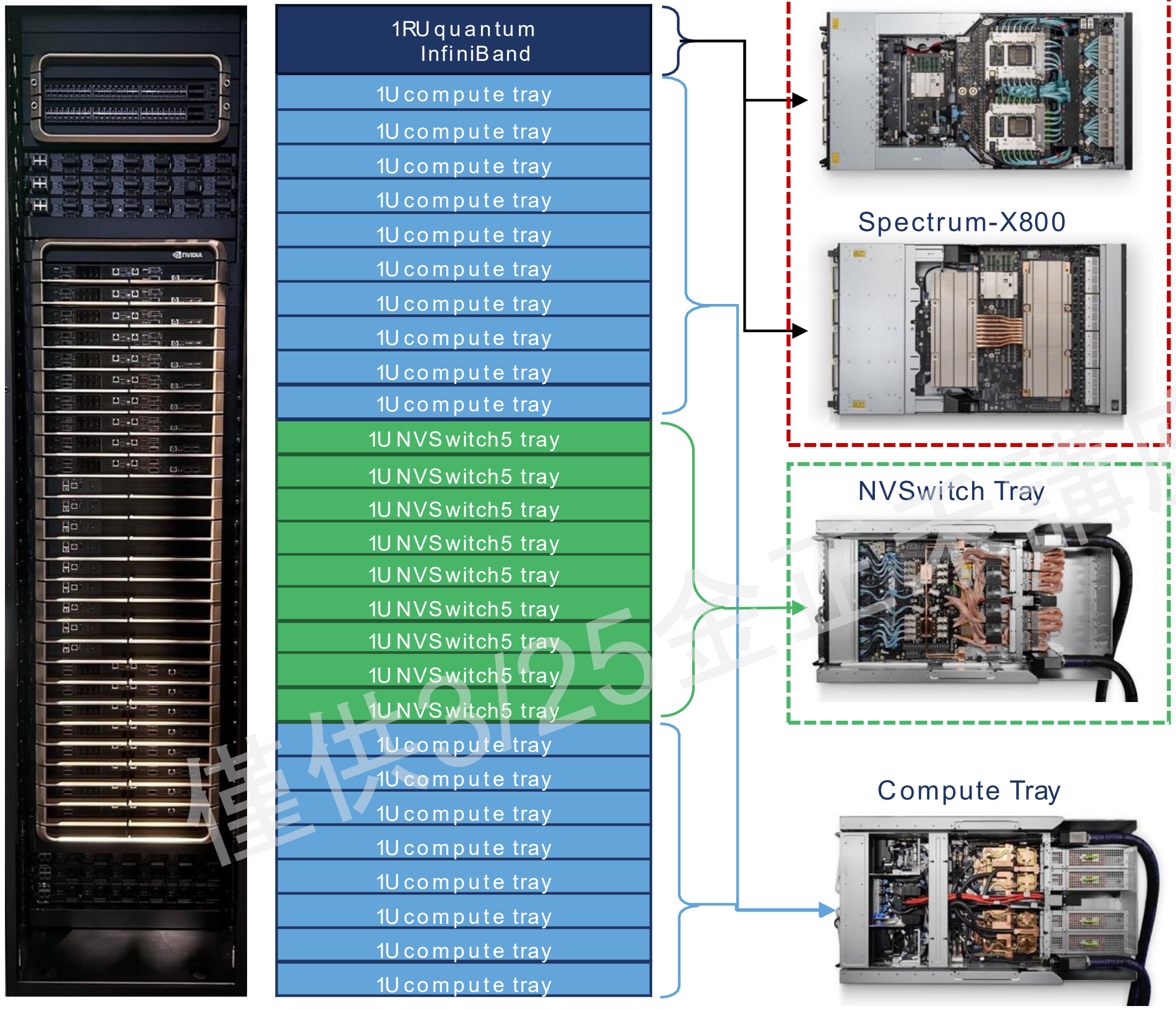
Minimize Latency Impact
Hide long-haul latency through topology, scheduling, and algorithm co-design for distributed AI.

"Break the bandwidth and latency wall beyond a single rack to build a truly global Big xPU."

Source: NVIDIA

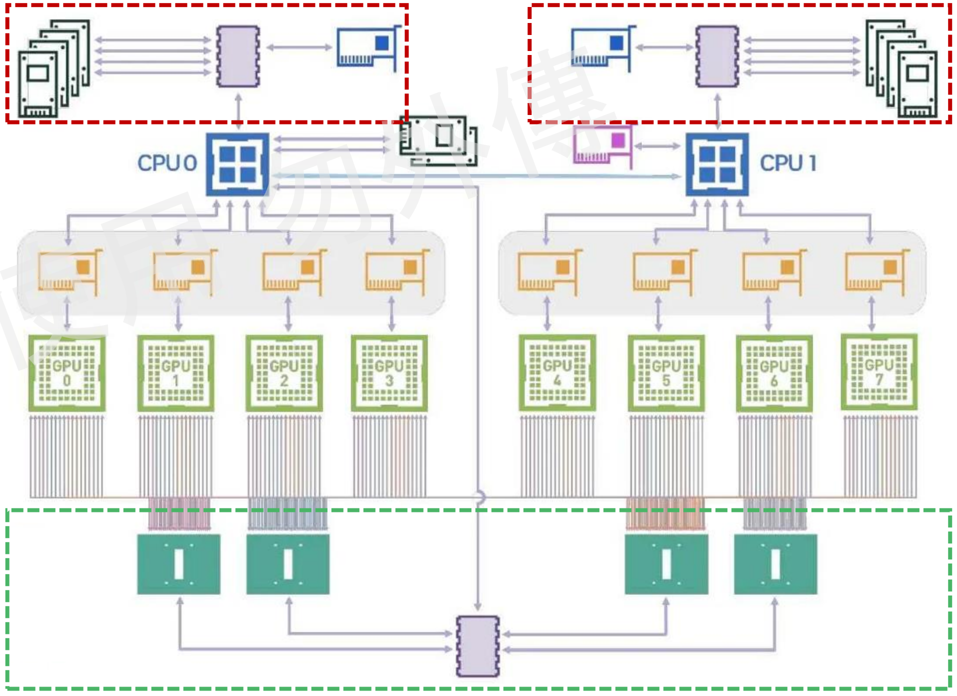
The Reality Check: Where is CPO Today?
Moving beyond Scale-Out switches to conquer Electrical Scale-Up.

NVIDIA DGX GB200 NVL72



To **Scale-Out** interconnection

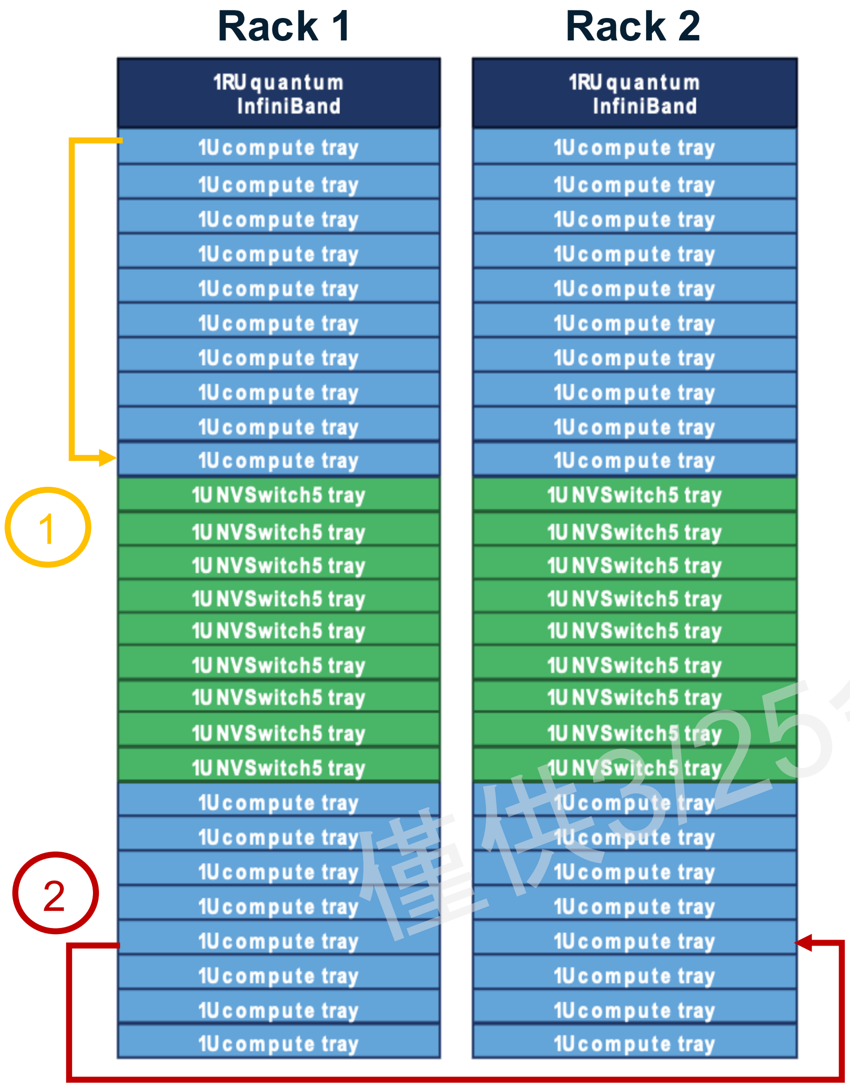
e.g. Quantum-X/Spectrum-X(Pluggable/CPO)



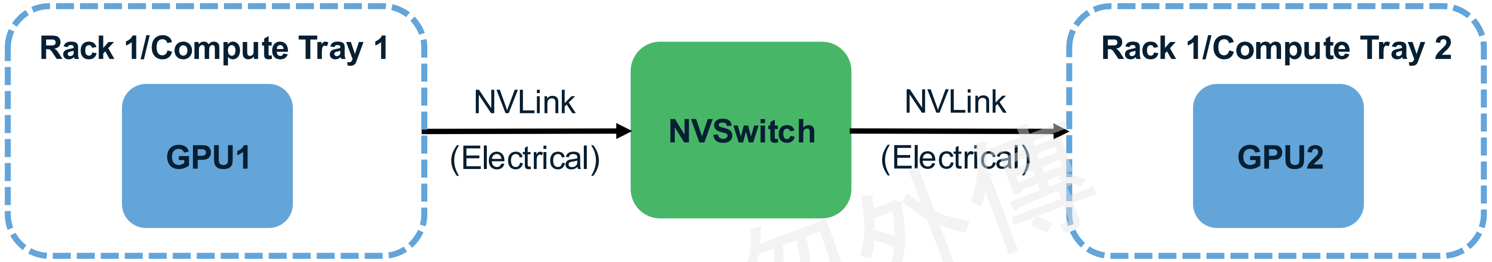
To **Scale-up** interconnection

e.g. NVSwitch(Electrical)

Difference Between Scale-up & Scale-out Interconnect
Extra protocol and O/E conversions degrade bandwidth and latency

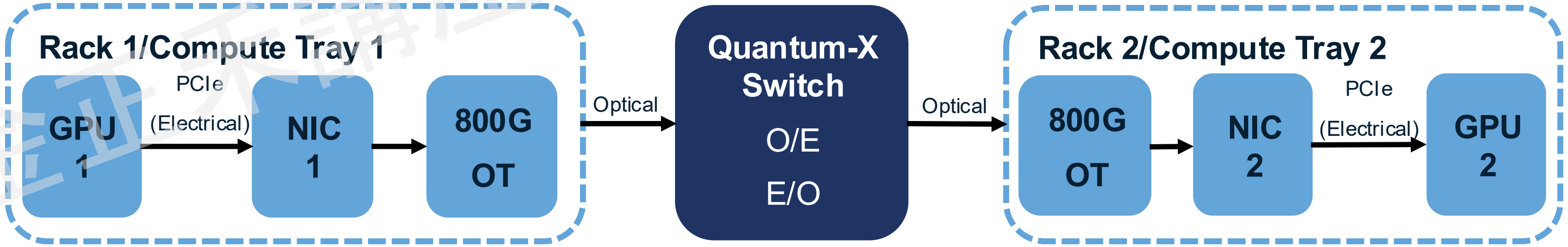


Scenario 1: Connections within same rack(Scale-up interconnect)



BW:7.2Tbps unidirectional per GPU Latency: ~100s of ns

Scenario 2: Connections across different rack(Scale-out interconnect)



BW:0.8Tbps unidirectional per GPU Latency: 2~3 of us

Scale-out architectures require multiple O↔E conversions, resulting in significantly degraded bandwidth and latency compared to scale-up, restricting computations to per-rack granularity.

Technologies for interconnection
Electrical based solutions will be limited while BW increase

Electrical: DAC, ACC, AEC
Optical: AOC, OT, CPO

		DAC Direct Attach Copper	ACC Active Copper Cable	AEC Active Electrical Cable	AOC Active Optical Cable	OT Optical Transceiver	CPO Co-Packaged Optics
Definition		Copper only with no active ICs	Copper with amplifiers	Copper with re-timer	Short reach fixed length transceivers	General pluggable optical transceiver	Transceivers integrated with xPU/Switch ASIC
Reach in different BW	100G /lane	3m	5m	7m	**Reach depends on design (at least 3m)** ➢ Tx/Rx ➢ Wavelength ➢ Modulation methods →Per lane BW will not compromise reach		
	200G /lane	2m	3m	4m			
	400G /lane	1.5m	2m	3m			
Power Consumption (pJ/bit)		0	3	5	20	15	<5
1.6T ASP($)		260	336	405	600	750	1000

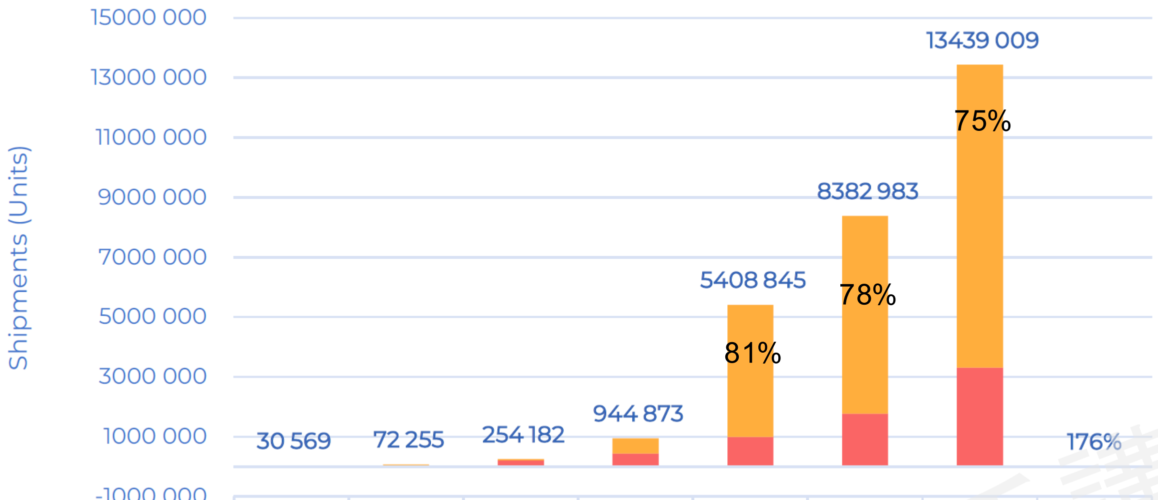
AI/DC needs at least >3m

- ➢ **As lane speeds scale toward 200G–400G, copper DAC/ACC/AEC links become increasingly reach-limited (typically <3 m), making optical interconnects the practical choice for extending compute connections beyond a rack.**

Source: Lightcounting/Semianalysis

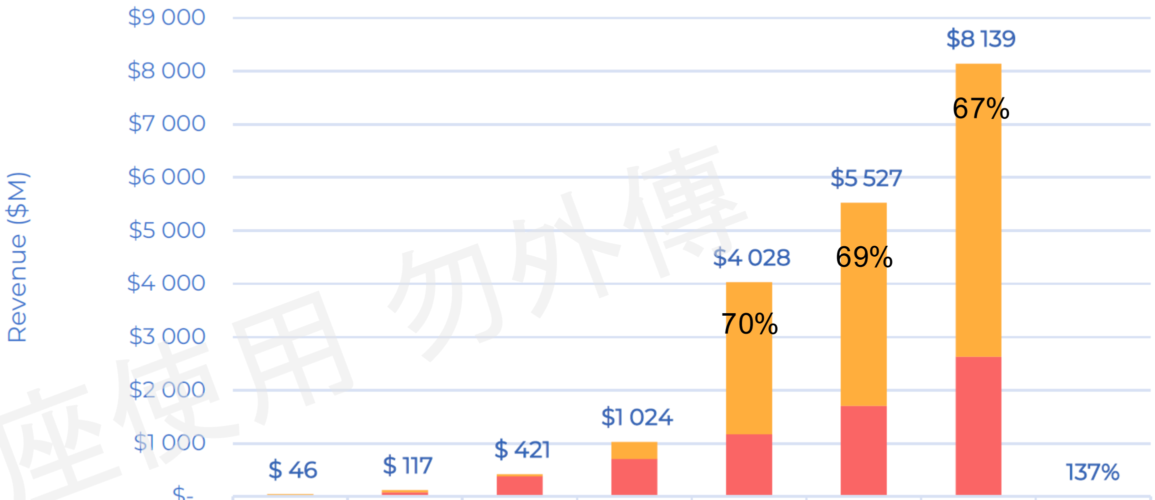
The TAM Validation from Yole: Scale-Up Dominates CPO Growth

2024-2030 Shipments of OEs fo Scale-out and Scale-up in Units



	2024	2025	2026	2027	2028	2029	2030	CAGR 24-30
TOTAL	30 569	72 255	254 182	944 873	5408 845	8382 983	13439 009	176%
TOTAL CPO Scale-Up	23 069	32 905	48 174	504 129	4413 825	6608 955	10124 009	176%
TOTAL CPO Scale-Out	7 500	39 349	206 008	440 743	995 019	1774 028	3315 000	176%

2024-2030 Revenue of OEs fo Scale-out and Scale-up in $M



	2024	2025	2026	2027	2028	2029	2030	CAGR 24-30
TOTAL	$ 46	$ 117	$ 421	$1 024	$4 028	$5 527	$8 139	137%
TOTAL CPO Scale-Up	$ 25	$ 38	$ 41	$ 315	$2 853	$3 820	$5 509	146%
TOTAL CPO Scale-Out	$ 21	$ 80	$ 380	$ 709	$1 176	$1 707	$2 630	124%

1. Scale-Out is Driven by High ASP; Scale-Up is Driven by Massive Volume

Insight: Scale-Out (inter-cluster connection) covers longer transmission distances (500m-2Km), requiring higher laser power, complex DSPs (Digital Signal Processors), and advanced error correction technologies. Consequently, it commands an ASP premium of approximately 15-20%.

2. The Learning Curve and Cost-Down Effect

Insight: From 2024 to 2030, the ASP for Scale-Up is projected to drop from $1,083 to $544 (a ~50% reduction). In the context of hardware infrastructure, this represents a highly healthy **"semiconductor economies of scale curve."**

3. The Shifting Supply Chain Moat

Insight: Since Scale-Up is fundamentally a game of "lower ASP, massive volumes, and high integration," traditional optical module manufacturers that rely on manual or semi-automated assembly to earn high margins (such as most legacy pluggable suppliers) will face severe headwinds.

Source: Yole

Technical Architecture and Implementation Paths for Scale-Up CPO

2

Winning the Scale-Up War:
Analyzing Next-Gen Interconnect Specs

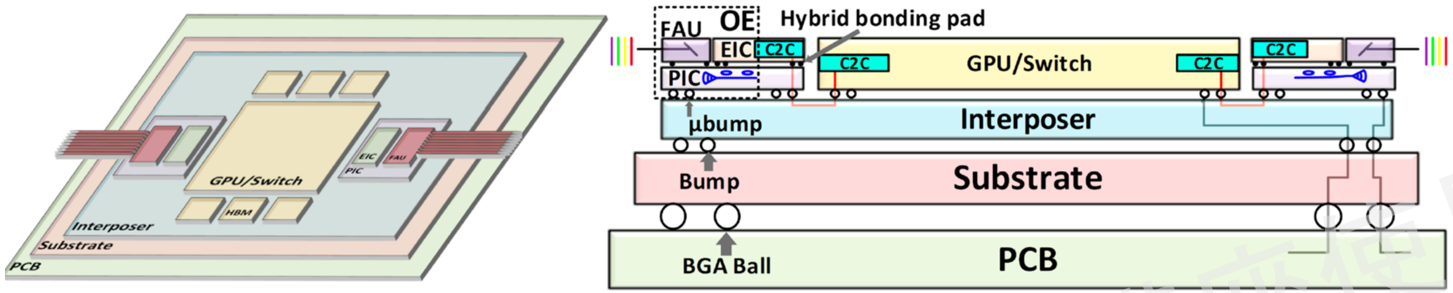
		Broadcom	NVIDIA	Ayar Labs	Ranovus	Celestial AI	Lightmatter
Solution Code		SCIP	Optical NVLink	TeraPHY	Odin	Photonic Fabric™	Passage™
Technology Architecture		Co-Packaged Optics (CPO)	Co-Packaged Optics (CPO)	Optical I/O Chiplet	Monolithic Silicon Photonics Engine	Photonic Fabric™ Interconnect Platform	3D Photonic Interposer
Per channel data rate		100Gbps	100Gbps	*32Gbps	100Gbps	*28Gbps	*56Gbps
# of channels		64	16	128	8	256	2048
OE Bandwidth Capacity		6.4Tbps	1.6Tbps	4Tbps	0.8Tbps	7.2 Tbps	114 Tbps
Coupling technology		**Edge Coupling (EC)**	**Grating Coupling (GC)**	**Grating Coupling (GC)**	**Edge Coupling (EC)**	**Grating Coupling (GC)**	**Grating Coupling (GC)**
ELS	**Module Type**	**Pluggable**	**Pluggable**	**Discrete module**	**X (integrated)**	**Pluggable**	**Discrete module**
	Wavelength	**1310nm**	**1310nm**	$16\lambda\ WDM$ **(O band)**	**1530nm-1565nm Comb**	**1310**	$8\lambda\ WDM$ **(O band)**
Energy Efficiency		Reduces Optical I/O power by 70%	Low-latency point-to-point optimization	4x to 8x more power efficient than traditional I/O	Ultra-low energy (<1 pJ/bit)	Reduces data movement energy by 60-90%	Supports >1.4 W/mm² power density

1. Fast & narrow/ slow & wide are two different approaches to realizing high BW interconnect spec.
2. GC may be the most important coupling methods since NV/Celestial AI/ Lightmatter adoption.
3. Most Scale-up CPO technologies rely on ELS, with monolithic integration (e.g., Ranovus) being a notable exception

*Notes: Solutions like Ayarlabs/Celectial AI/Light matter called Slow & Wide interconnection

The Shift to Scale-Up CPO:
Anatomy of NVIDIA’s "Optics on Interposer"

NV’s Optics on Interposer structure for Scale-Up



1. Optics as a Packaging Primitive: The 3D-Stacked Era

➢ NVIDIA is transitioning from "connecting chips" to "integrating optics" directly into the 3D package.

2. Shattering the "Shoreline Bottleneck" via Vertical Coupling

➢ The architecture is engineered specifically to match the extreme Shoreline Density requirements of next-gen GPUs.

3. Reach Extension without the "Latency Tax"

➢ This design extends the reach of C2C-USR (Ultra-Short Reach) links while maintaining on-chip latency performance.

Optical component ratio per GPU

Item	Ratio	Description
GPU : CPO OE	1 : 2	2 OEs per GPU for NVLink scaling.
GPU : FAU	1 : 2	1:1 mapping with OEs.
GPU : ELS	1 : 0.5	1 ELS shared by 2 GPUs (4 OEs).

Calculation for per Tray (4 GPUs) optics usage

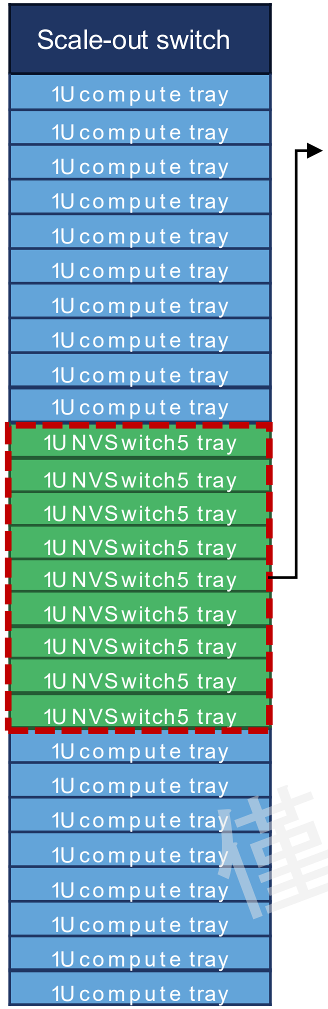
Item	Usage
CPO OE	8
FAU	8
ELS	2

All these amount are new to the tray since Scale-up CPO adopted

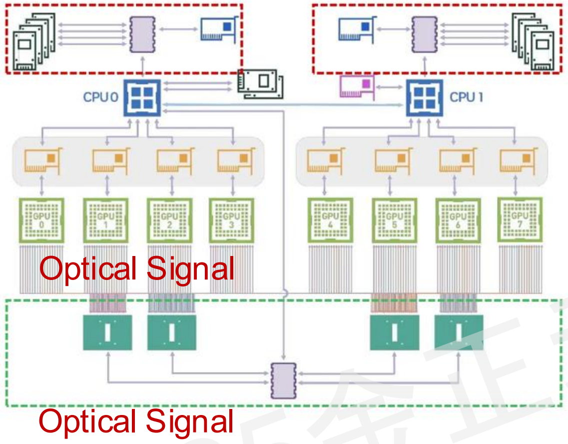
Assumption for calculation: Scale-up OE remains identical to the scale-out architecture; compute trays follow the GB200 design (4 GPUs per tray).

Source: NVIDIA ISSCC 2026

The 5x Growth Engine: Mapping the "Full-Photonics" Rack Opportunity
Scaling from 72 to 360 OEs through full-rack CPO integration in NVL72



NVSwitch optics usage



Scale-up switch side need same number of optics,
→Optics number x2

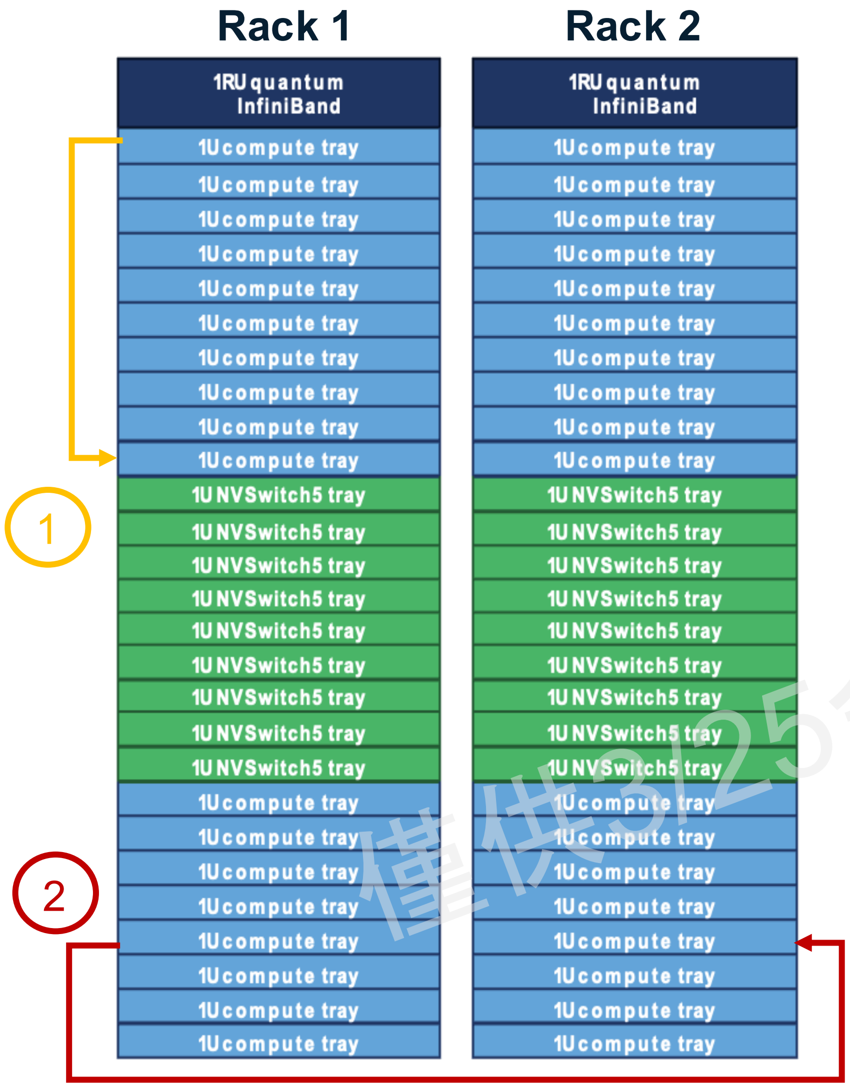
Rack level sum of optics usage

Total Component Count		Scenario A: Pure Scale-Out CPO (Scale-Out only)	Scenario B: Full CPO (Scale-Up + Scale-Out)
Inside Scale-out Switch	# OE	72 OE	
	# FAU	72 FAU	
	# ELS	18 ELS	
Inside Compute Tray	# OE	-	8OE/tray*18tray =144 OE
	# FAU	-	8FAU/tray*18tray =144 FAU
	# ELS	-	144OE/4 OE pre ELS= 36ELS
Inside Scale-up Switch	# OE	-	144 OE
	# FAU	-	144 FAU
	# ELS	-	36ELS
Per Rack OE		72	360
Per Rack FAU		72	360
Per Rack ELS		18	90

Scale-up switch side need same number of optics,

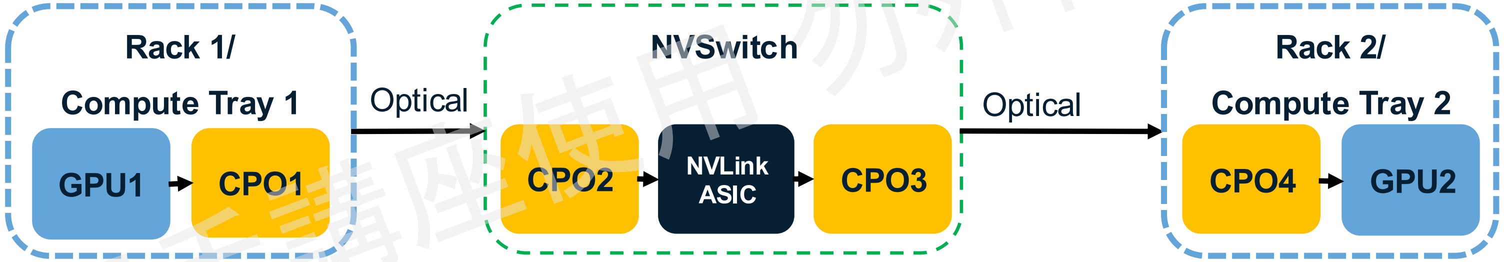
Assumption for calculation: Use NVLink 5 for scale-up connection

Unified CPO Fabric: The Future of Scalable AI Infrastructure
Streamlined Connectivity with OEs for Massive BW Density



Scenario 1 : Scale-up interconnect with NVSwitch(CPO based, intra rack)

Scenario 2 : Scale-up interconnect with NVSwitch(CPO based, inter rack)*



BW:1.6-3.2Tbps bi-direction per GPU Latency: ~100-200 of nS

Optical Scale-up fabric unlocks massive GPU parallelism by providing unprecedented bandwidth and ultra-low latency, enabling a larger number of GPUs to collaborate more efficiently than ever before.

*Notes: Assume optical signal can extend the reach @ Scenario 2

The Economics of Scale-Up CPO: Supply Chain Value Reconstruction

3

CPO supply chain:
Current Scale-Out CPO supply chain

PIC design	Substrates/Lasers/ Connectors/Fibers/ Others	Foundry	ASIC/xPU partners	OSATs/ SiP packaging	CPO equipment	Customer
NVIDIA BROADCOM MARVELL CISCO	*Epi-wafers* LandMark VPEC *Lasers* LUMENTUM COHERENT BROADCOM *Fibers* CORNING SUMITOMO ELECTRIC *FAU* SENKO CORNING molex Teramount Amphenol FOCI phix	*CPO PICs* tsmc GlobalFoundries SILTERRA *CPO ASICs* tsmc	NVIDIA, CISCO *SpectrumX ASIC* *QuantumX ASIC* BROADCOM *CPO ASIC TH5* MARVELL *Teralynx ASIC* CISCO, NVIDIA AMD	*Packaging* ASE HOLDINGS SPIL *Testing* iST Integrated Service Technology *System assembling* Foxconn 鴻海科技集團 Fii Foxconn Industrial Internet fabrinet *SiP packaging* ShunSin *FAU assembly* fabrinet	NVIDIA Micas Networks DELTA Juniper Networks Ruijie Networks 锐捷 H3C CISCO Celestica	aws Microsoft Azure Meta TESLA xAI VULTR ORACLE black forest labs MISTRAL AI_ Tencent 腾讯 ByteDance Alibaba Group 阿里巴巴集团 Baidu 百度
RANOVUS	NRC · CNRC Laser	GlobalFoundries CPO PIC	MEDIATEK CPO ASIC			

Source: Yole 2025

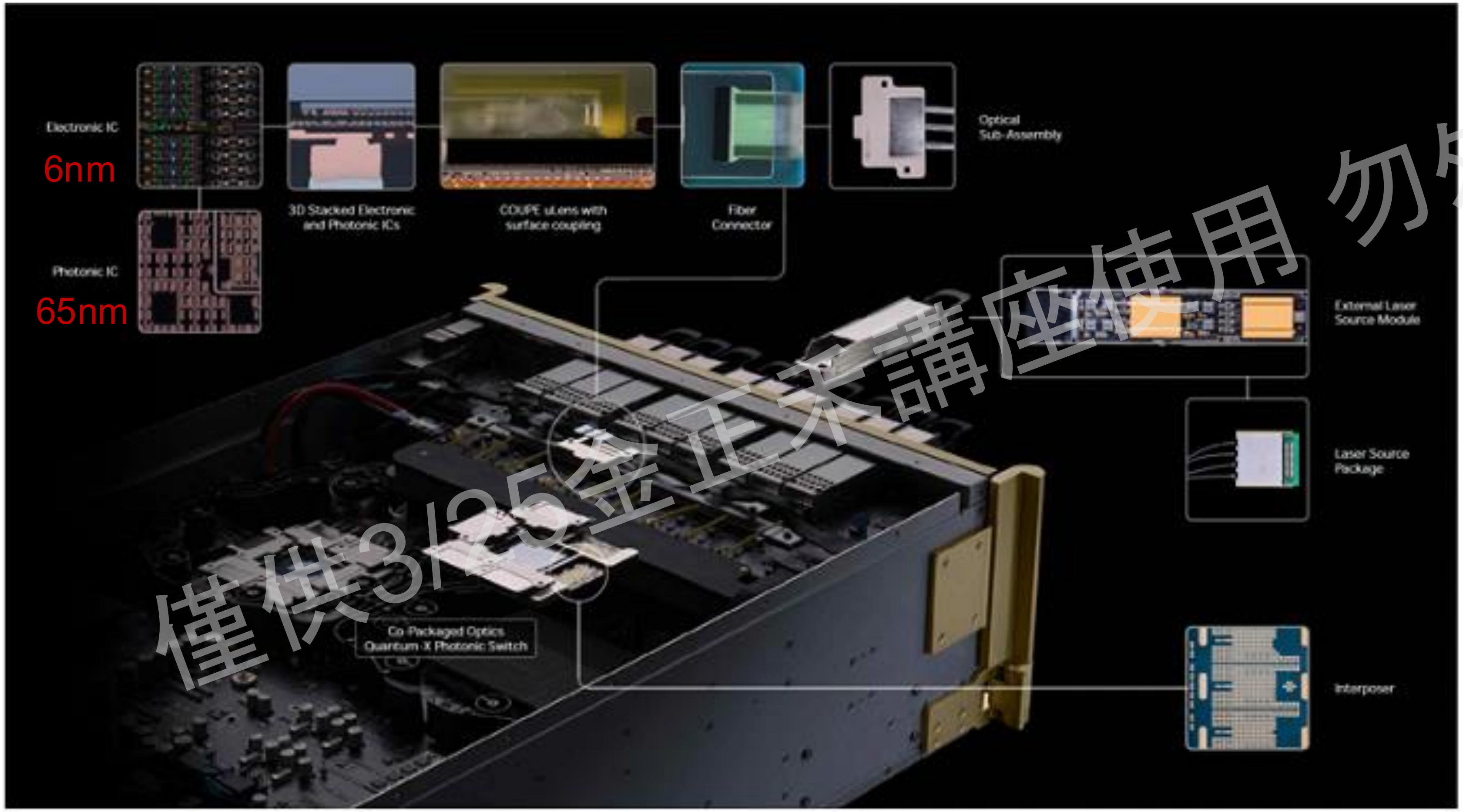
Supply Chain Continuity: Bridging Scale-Out and Scale-Up CPO

tsmc — EIC/PIC/COUPE

OSAT: SPIL, ASE

FAU: TFC 天孚通信, SUMITOMO ELECTRIC, SENKO, FOCI

FAU assembly: FOXCONN, fabrinet



ELS: SUMITOMO ELECTRIC, TFC 天孚通信, fabrinet, LUMENTUM, COHERENT, 波若威科技 BROWAVE CORPORATION, O-Net Technologies, FURUKAWA ELECTRIC

Connector: T&S, Amphenol, samtec, SENKO, SIEMON

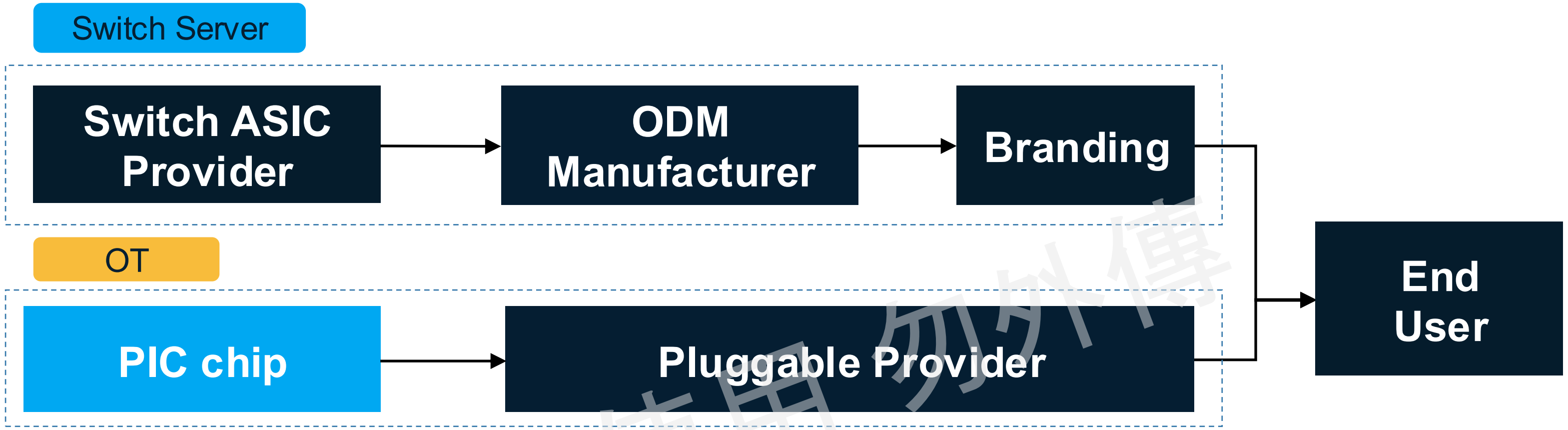
Fiber: CORNING, YOFC, HTGD

Source: SEMIVISION

Supply Chain Transformation: From Pluggable to CPO
The Shift from Component Assembly to System Integration

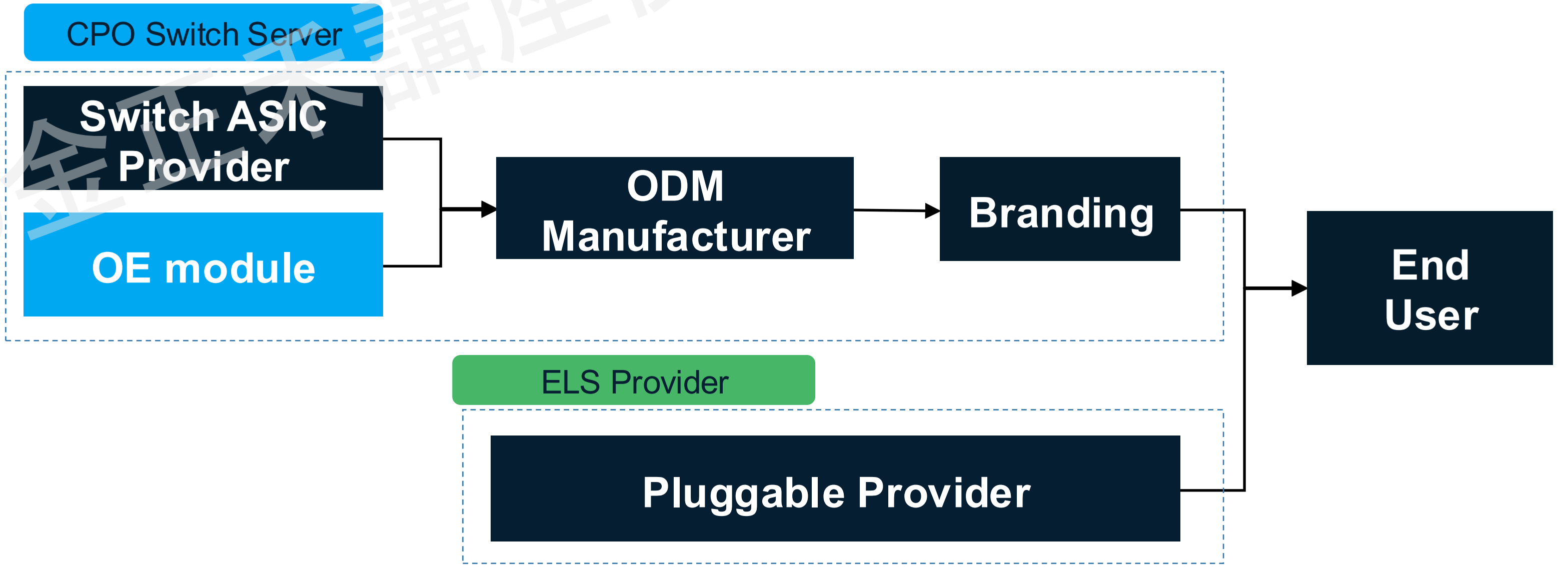
Conventional switch server with SiPh OT:

1. Optical and electrical components are completely decoupled
2. No advanced process needed
3. Precision assembly is concentrated(Pluggable provider)

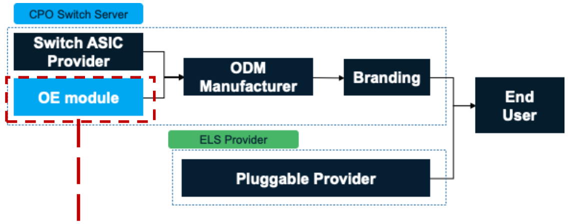


CPO switch server with ELS:

1. 1. Optics are integrated/embedded into the Switch server architecture
2. 2. Requires advanced packaging processes(EPIC couping)
3. 3. Precision assembly is dispersed(OE/ ELS provider)

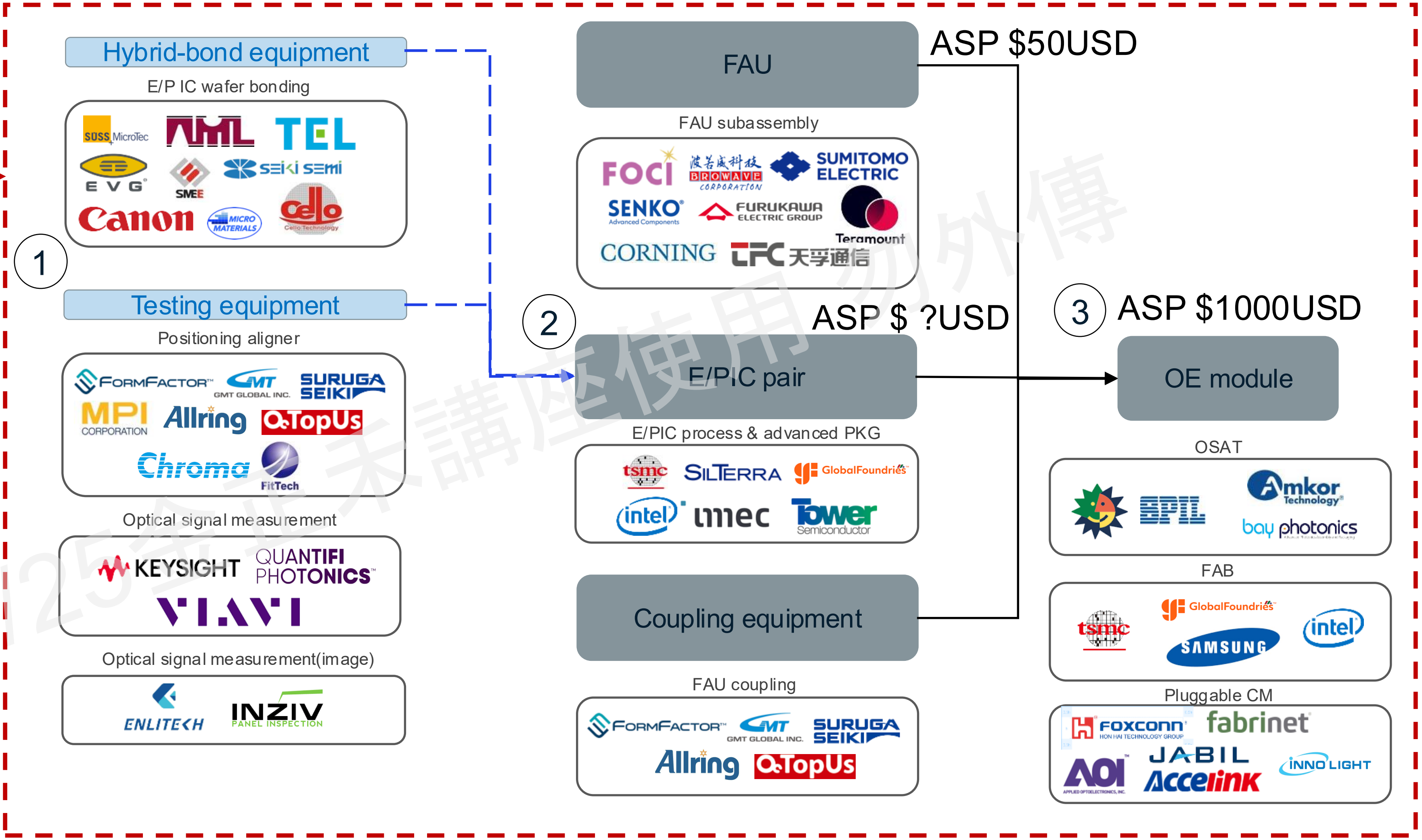


The Convergence of Silicon Photonics: A Supply Chain Power Shift Competition Between Fabs, OSATs, and Traditional Manufacturers

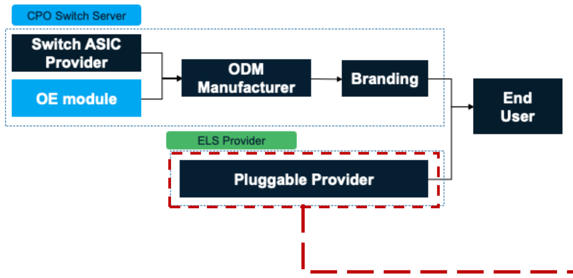


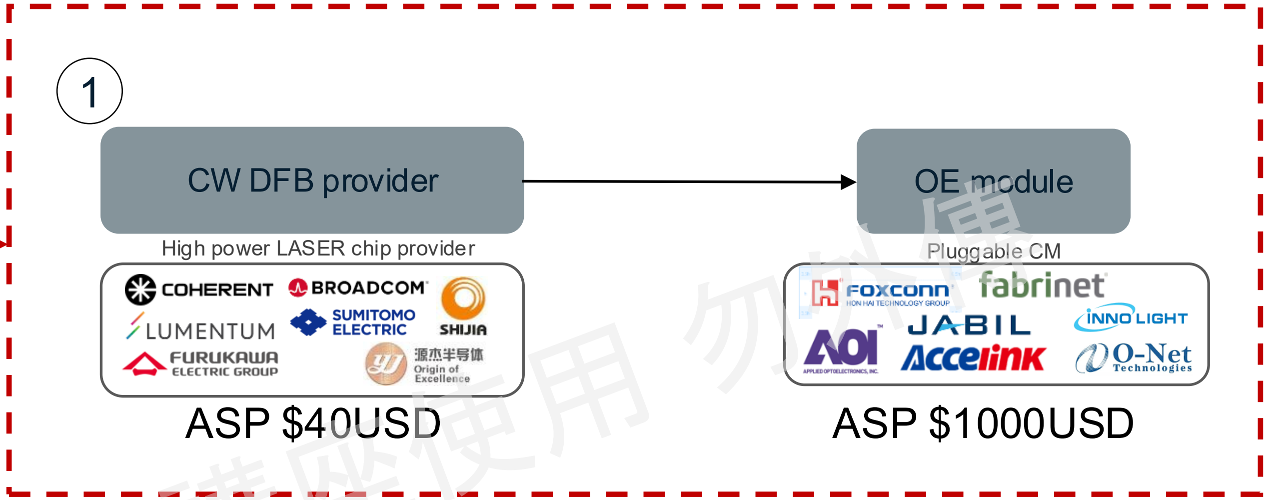
1.**E/PIC pair** requires **wafer-to-wafer/hybrid bonding**, increasing costs for additional bonding equipment and testing.
2.Bonded **E/PIC pairs** are **optoelectronically coupled**, posing several measurement challenges:
- ✓ How to apply probes for EIC electrical signals to control the PIC.
- ✓ How to couple CW light into the PIC.
- ✓ How to couple out and analyze the modulated optical signals.

3.For **OE modules**, beyond traditional **pluggable CMs**, **OSATs** and **Fabs** may also seek to perform their own OE module assembly.

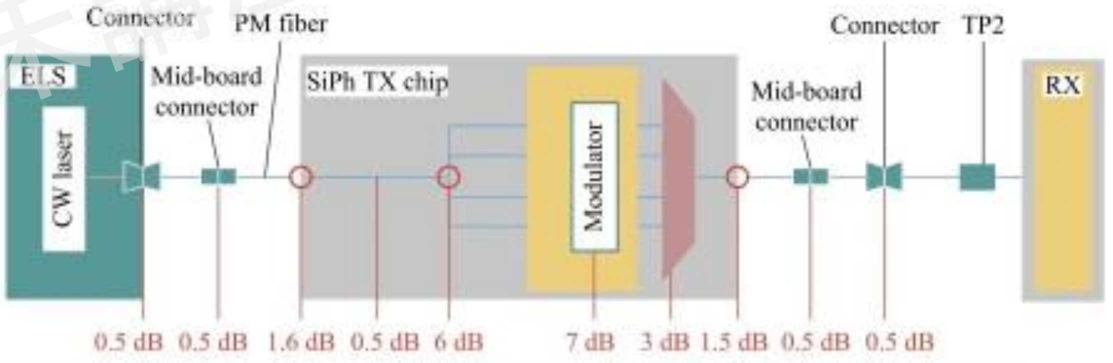


CPO Supply Chain Dynamics: The Evolution of ELS Requirements High Specs/uniformity as a Barrier to Entry





1. ELS must provide sufficiently strong light to ensure that the light can traverse the entire link and be received by the Rx in another CPO.
2. Currently, the DFB laser power used in the single 1310nm wavelength is at least 100mW. At this wavelength and light intensity, Chinese manufacturers are still barely able to provide ELS light sources.
3. In the future, if the requirement moves toward 200mW/DFB or even 400mW/DFB @ 1310nm, or uses multi-wavelength CWDM (1270/1290/1310/1330/1350nm), Chinese manufacturers may temporarily be unable to meet the demands in terms of intensity and multi-wavelength uniformity.



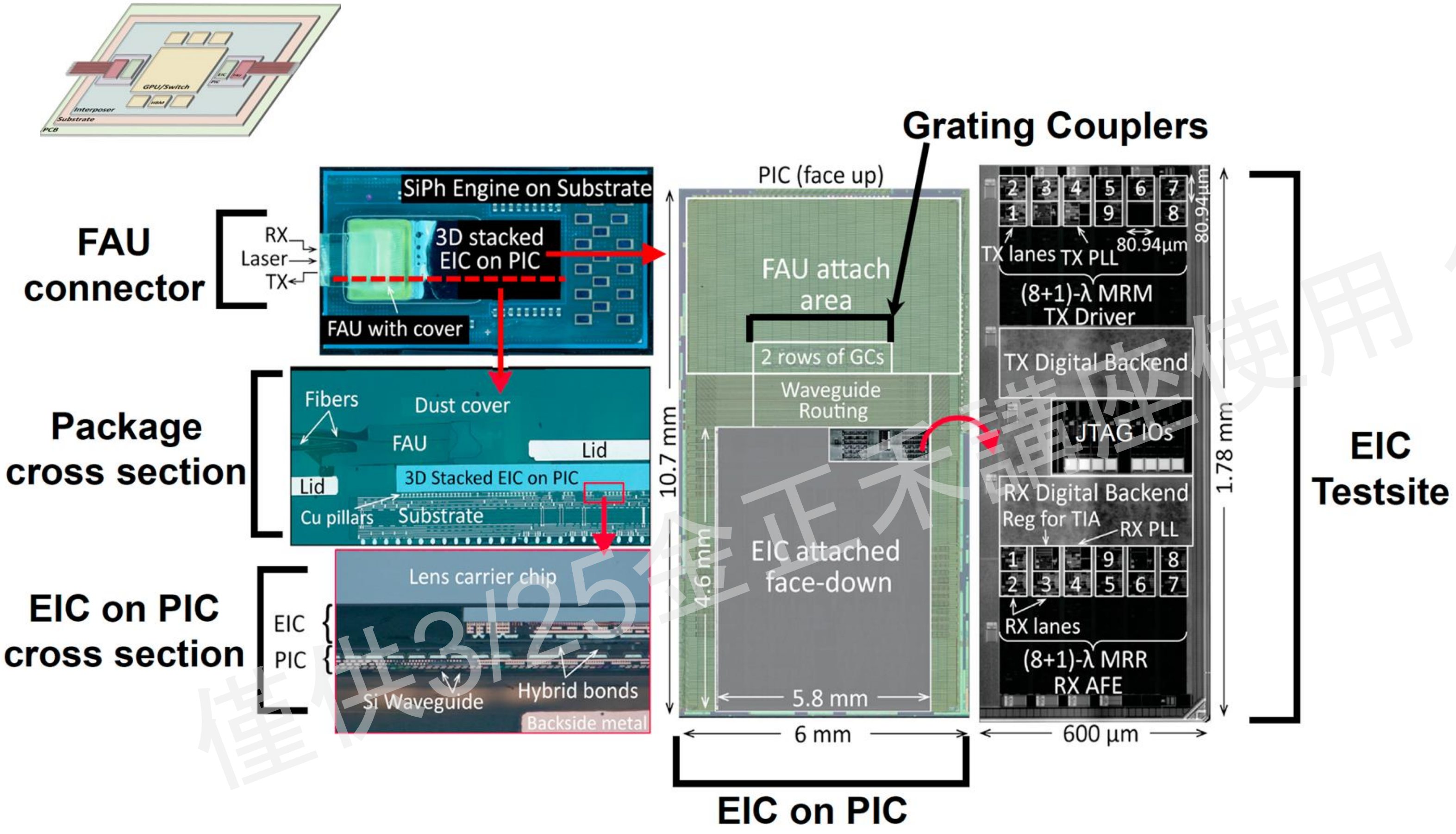
Power budget in the whole CPO link

→ELS need to provide enough power for each CPO channel

Technical and Commercial Risk Assessment of Scale-Up CPO

4

NV Scale-Up CPO OE packaging structure
MRM x Hybrid Bonding x Grating Coupling



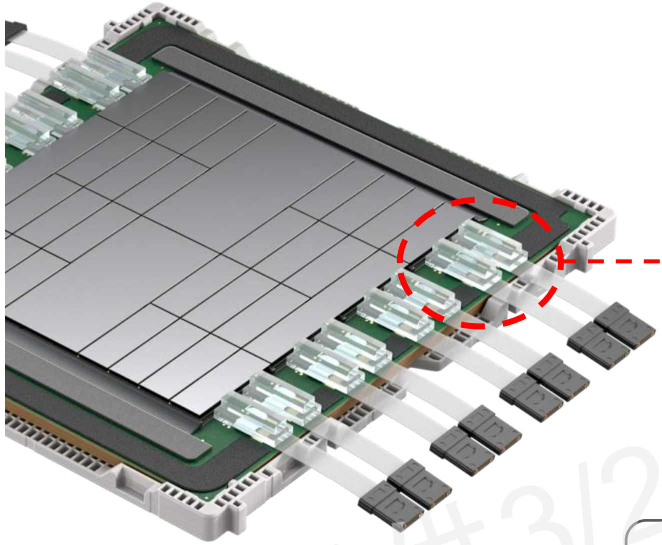
1. Extreme Footprint Efficiency: The 64.2mm^2 Solution
Insight: NVIDIA achieves unparalleled density by 3D-stacking the Electronic IC (EIC) and Photonic IC (PIC) into a compact 6mm x 10.7mm footprint.

2. Inherited MRM Architecture: WDM Integration for High-Density I/O
Insight: The engine utilizes (8+1)lambda Microring Modulators (MRM), a high-efficiency technology derived from NVIDIA's proven Scale-out CPO networking architecture.

3. Surgical Precision: Transitioning to Advanced Semiconductor Packaging
Insight: The layout features two rows of Grating Couplers (GCs) and a dedicated FAU attach area, signaling a shift from optical assembly to semiconductor-grade precision.

Source: NVIDIA

FAU Challenges in CPO: Bottlenecks & Future Trends
Why DFAU is Mandatory for Future High-Density Interconnects.



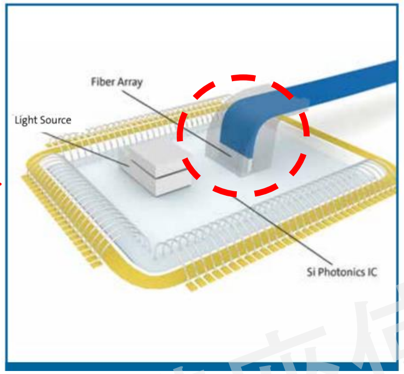
xPU/HMB with CPO OE

Detachable socket

Socket for FAU

LOTES Amphenol® TE molex® SENKO

FAUs: Coupling & guiding light into fiber

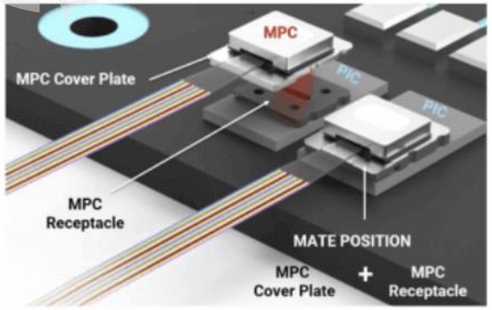


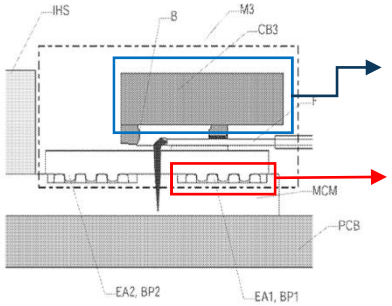
FAU assembly example for grating coupler and redirect the light physically by fiber bending .

→The FAUs will break while system assembly

(Post OE module process)

Detachable-FAUs(DFAU): FAU inside reworkable socket





Socket for FAU

u-Structure for alignment assistance

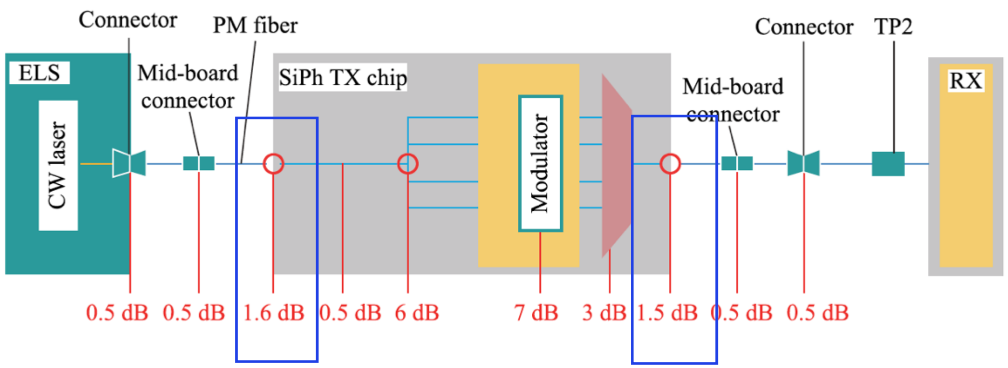
(Normally Si based)

1. FAU + reworkable socket with AA coupling equipment
2. Replace fiber array with new one if broken

Source: Lightmatter,SENKO

Is Active Alignment (AA) Mandatory for CPO OEs? PA May Fails to Meet the Power Budget While FAU assembly.

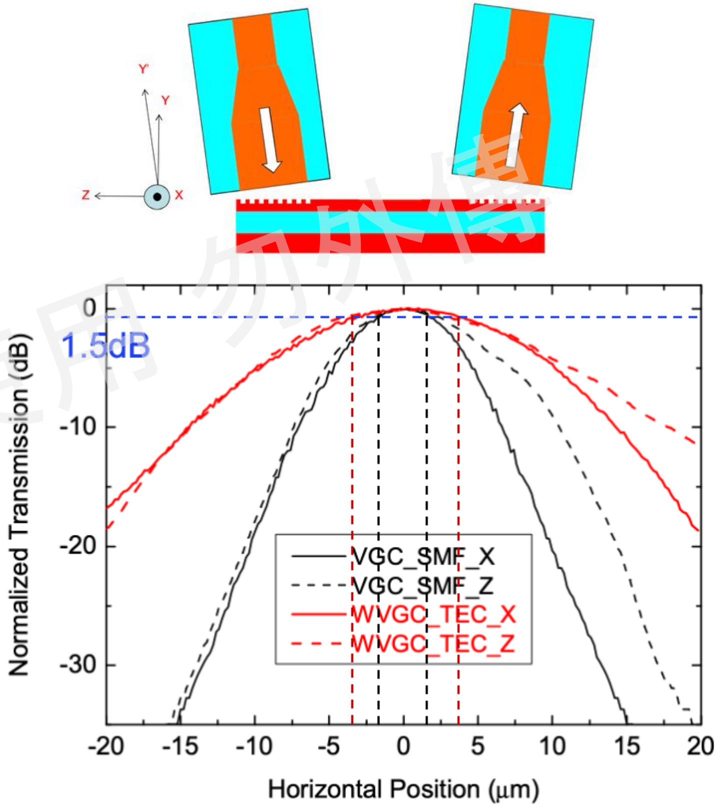
Power budget requirement in CPO system



Require FAU assembly

- The ELS output must cover the total link loss (~24.5 dBm here; 100 mW/channel) to ensure sufficient power reaches the Rx for O/E conversion and demodulation.
- The PIC transmitter has two array interfaces that require an FAU for optical coupling in and out.
- Current AA alignment for E/PIC and FAU takes 6-10 minutes. ➔ Throughput is limited to 6-10 CPO OEs per hour per machine.

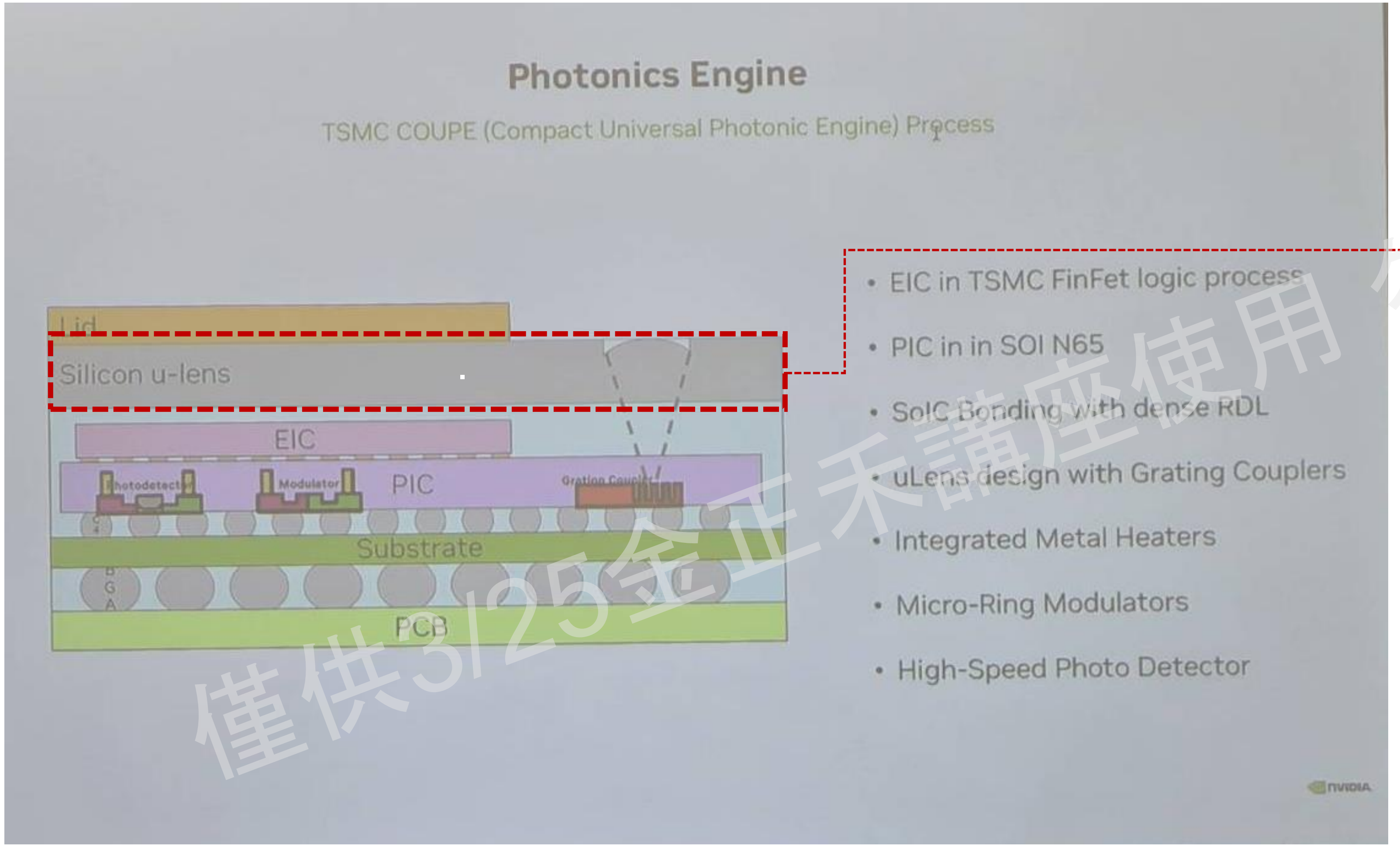
Relation between alignment position and loss



- Need < $\pm 5um$ assembly accuracy for a 1.5dB loss
- Fiber array also have tolerance, use PA may let several channel's loss >24.5dBm→ Rx can't read signal

Source: Co-packaged optics (CPO): status, challenges, and solutions(Springer, 2023)
Enhancing alignment tolerance of silicon waveguide by using a wide grating coupler(OPTICS EXPRESS, 2008)

Package Structure of COUPE Photonics Engine
Silicon u-lens help to enlarge the tolerance of FAUs



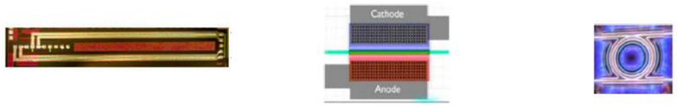
- ✓ Silicon u-lens on grating coupler help to enlarge the tolerance of FAU itself.
- ✓ AA coupling help to assembly the whole array to right place.

Source: NVIDIA 2026 OFC

Investment Opportunities and Business Outlook

5

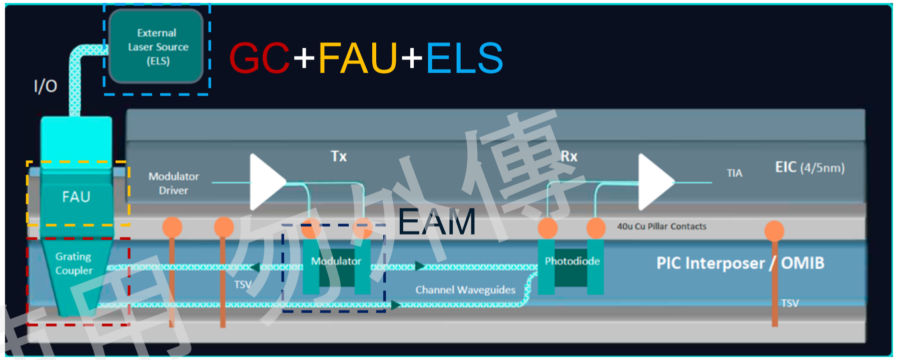
Diversified Modulator Solutions and CPO Integration Architectures
Diverse Modulator Paths, Same Optical Interface (ELS+GC+FAU)



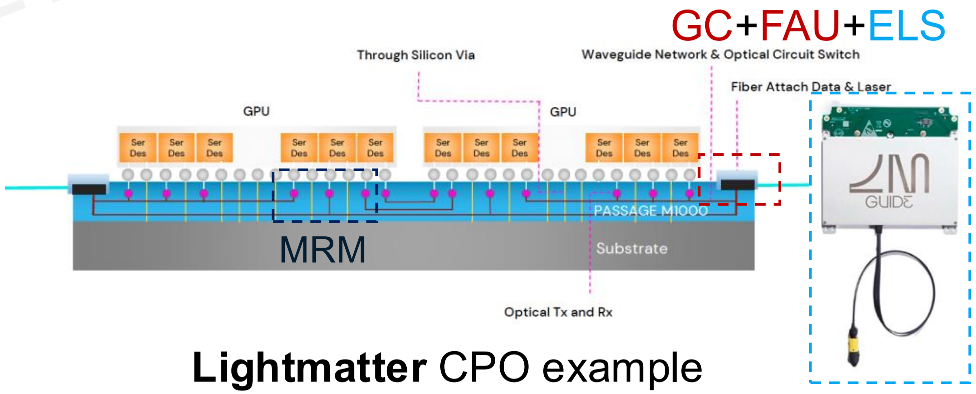
	MZM	EAM	MRM
Size	1000-1200um	~50-100um	~15um
Thermal Stability	>50°C	>50°C	<1°C
Speed(Data Rate)	>56Gbps	>56Gbps	>56Gbps
Power consumption	50mW	10mW	1mW
Transmitter penalty	<5dB	<10dB	<5dB
Need additional multiplexer?	Yes	Yes	No
Optical bands	O/C band	C-band only	O/C band
Who use?	Conventional SiPh OT	Celestial AI	NVIDIA/Ayarlabs/ Lightmatter

1. MZM too large, will reduce the data rate density.
2. EAM is balanced in both size, thermal stability & power consumption, Celestial has been acquired by Marvell.
3. MRM has the smelliest size and lowest power consumption but need additional feedback control for high temperature operation(optical interposer)

What will promised in optical interposer-based CPO?



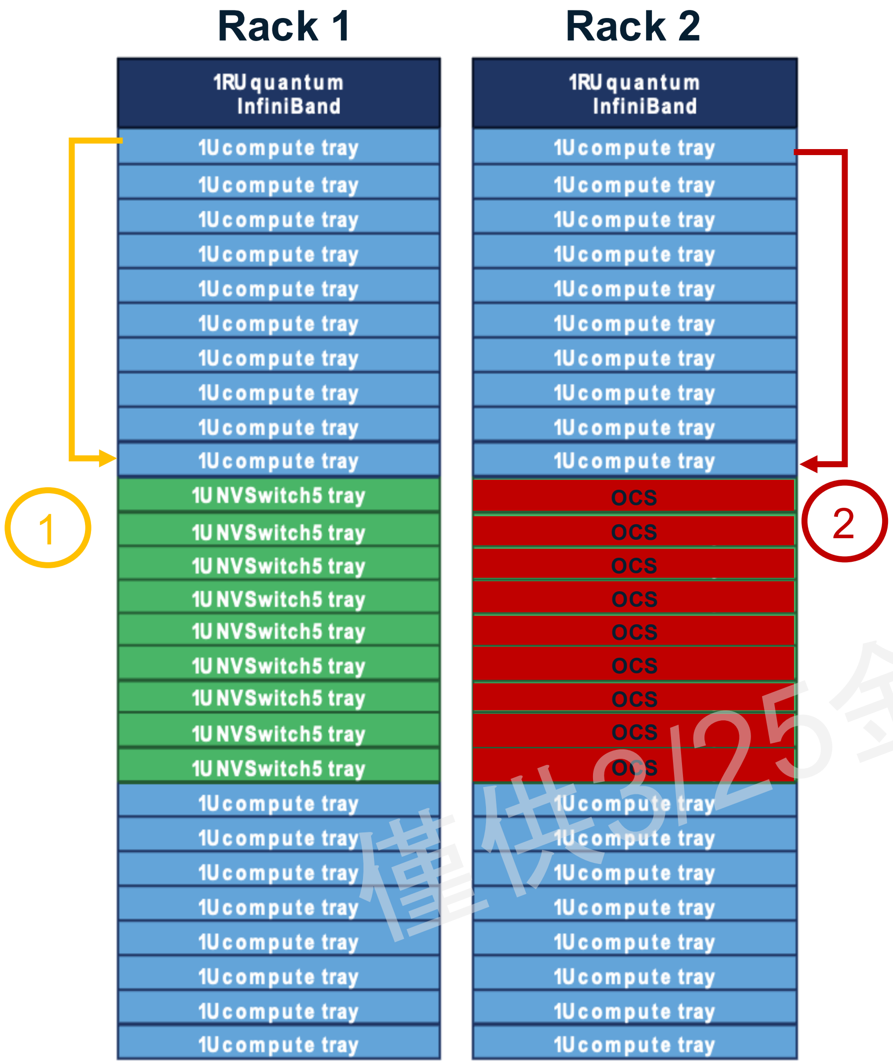
Celestial AI(part of Marvell) CPO example



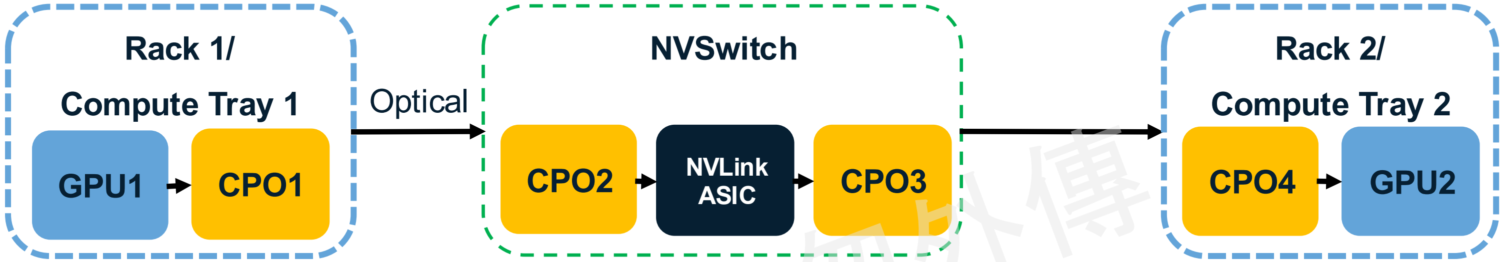
Lightmatter CPO example

Regardless of modulator selection or whether an interposer-based CPO is used, the integration of GC, FAU, and ELS remains indispensable.

Beyond CPO: The Quest for Zero-Conversion Optical Fabrics
Moving from O-E-O to O-O-O Efficiency with OCS Architectures



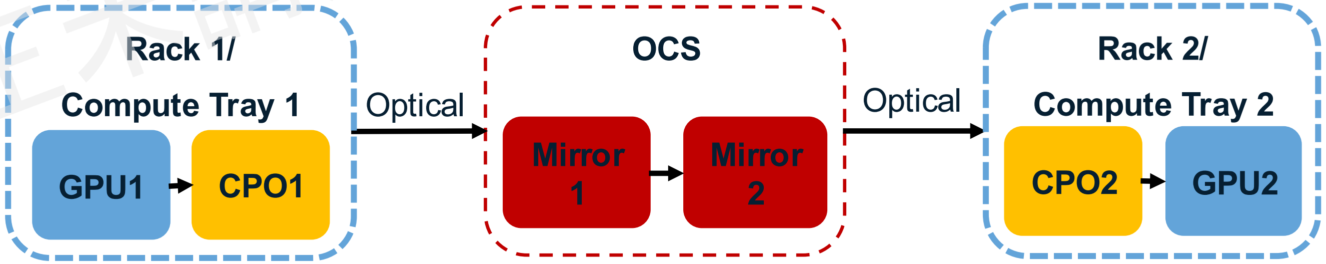
Scenario 1 : Scale-up interconnect with NVSwitch(CPO based, intra rack)



BW:1.6-3.2Tbps bi-direction per GPU Latency: ~100-200 of nS

Power consumption: 100-200pJ/bit(system level)

Scenario 2 : Scale-up interconnect with OCS(CPO based, inter rack)



BW:1.6-3.2Tbps bi-direction per GPU Latency: ~100 of nS

Power consumption: 20-30pJ/bit(system level)

Different Technologies for OCS
Ports, Loss & Speed Are the main spec. for OCS

Google BRIGHT Silicon Technologies LUMENTUM DiCon FIBEROPTICS | COHERENT | TELESCENT | HUBER+SUHNER | iPRONICS n-eye

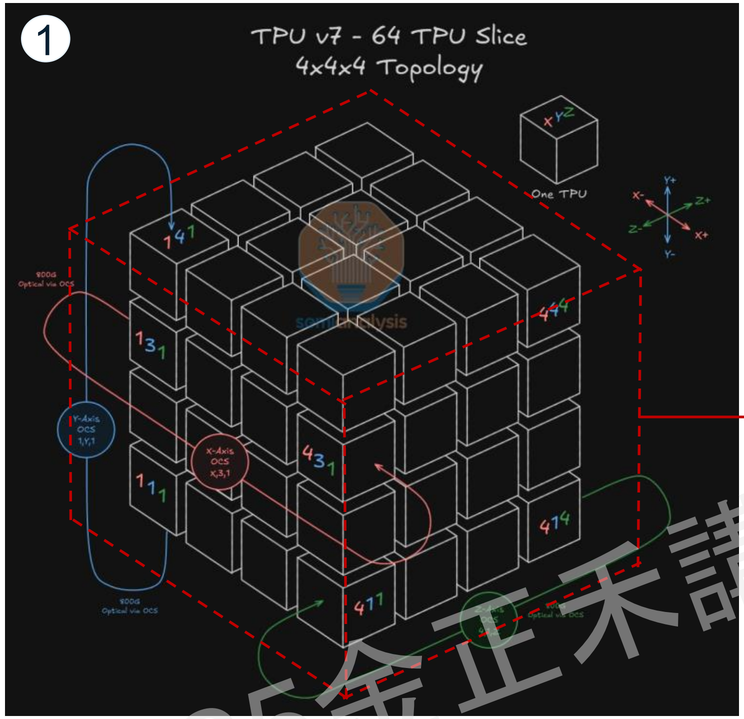
Technology	MEMS	Liquid Crystal	Robotic	Piezoelectric	SiPho
Methods	Utilizing micro-mechanical forces to actuate mirrors for precise beam steering within free space.	Manipulating the refractive index of liquid crystal layers via voltage to create phase gradients, enabling non-mechanical beam steering through optical diffraction	Utilizing high-precision robotic arms to physically reconfigure optical fiber connections through automated pick-and-place or alignment mechanisms	Harnessing the piezoelectric effect to generate precise micro-deformations in materials, driving the physical displacement of fibers or mirrors for optical switching	Routing light through integrated silicon waveguides and utilizing Mach-Zehnder Interferometers (MZI) to switch optical paths via thermo-optic or electro-optic phase shifting
Port count	600	512	1000	576	>1000
Insertion loss	<1.5dB	<3dB	<0.5dB	<1.5dB	TBD
Switch speed	mS	mS	S	mS	uS
Applications					
Spine Layer Replacement (Seldom lane change)	Google Deployed	Good	Ideal	Good	No
AI Cluster Reconfiguration (Medium lane change)	Google Deployed	Good	OK	Good	Good
AI Back-End (Frequently lane change)	OK	OK	No	OK	Ideal

Currently there does not exist a perfect technology solution for different application scenarios, a high port count, low loss & high switch speed will always the holy grail for OCS system.

Where is The Future?
A Google Case Study: Topology Reconfiguration via OCS

1. Each TPU can connect with adjacent TPU by copper or connect to the opposite face by optical and OCS.
2. Use 64 TPU block as a basic unit, a 4096/9216 TPU slice can be create by connect “+” faces to “-” faces with OCS.
3. With this topology, the Scale-up capital cost is only 40% compared to GB300 NVL72 solution with lower power consumption by utilizing OCS.

1



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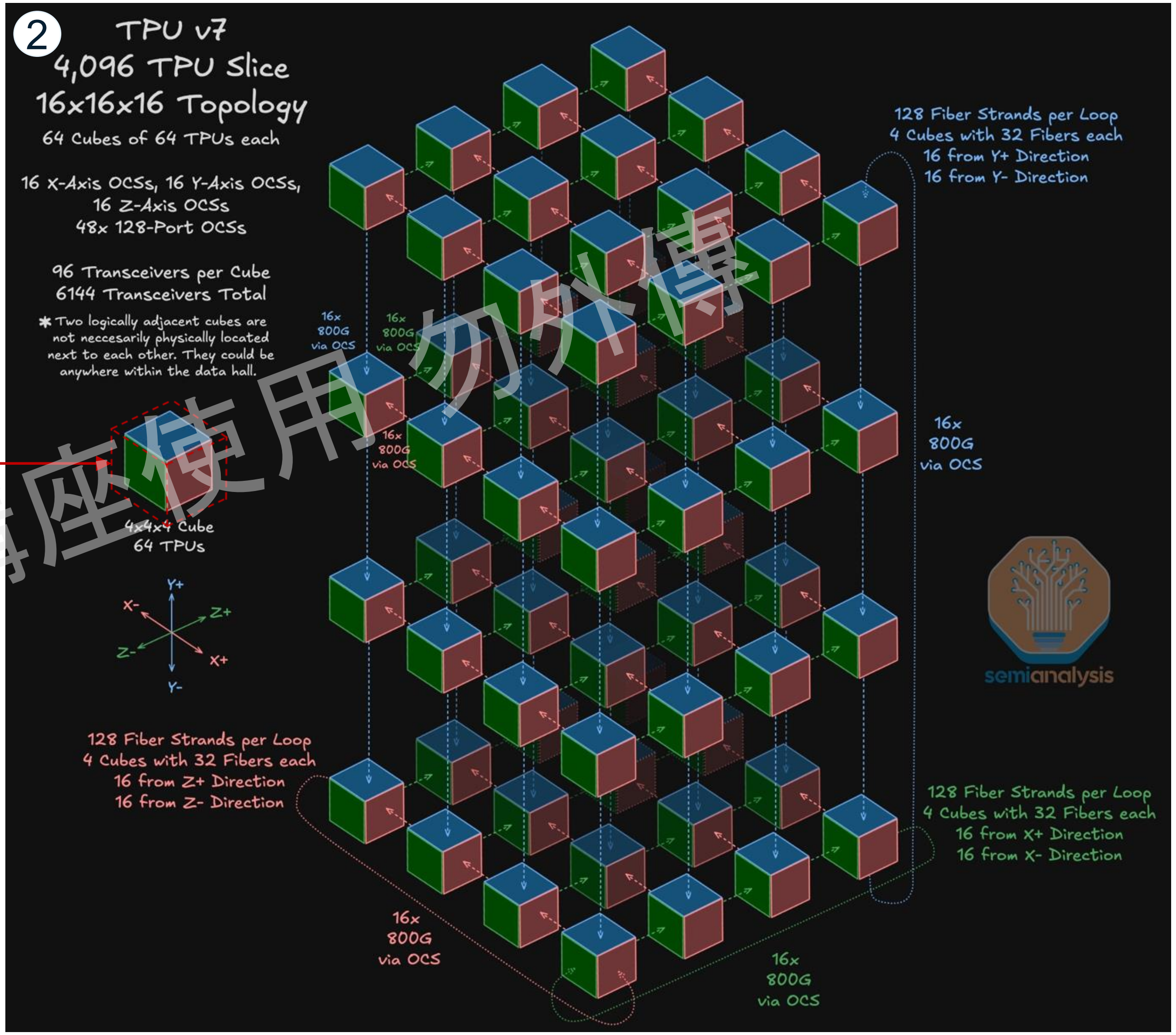

3

Scale-Up Network Cost Comparison	TPUv7 ICI	Trn2 Teton 2 PDS	GB300 NVL72
Scale-Up Networking Capital Cost			
Transceivers	[redacted]	[redacted]	[redacted]
Copper Backplane	[redacted]	[redacted]	[redacted]
Copper Cables (ex-PCB)	[redacted]	[redacted]	[redacted]
Switches	[redacted]	[redacted]	[redacted]
Fiber and Others	[redacted]	[redacted]	[redacted]
Total Scale-Up Capital Cost (9k XPUs)	**$17,091,840**	**$16,144,026**	**$42,297,446**
per XPU	$1,855	$1,752	$4,590
Cluster Size	9,216	9,216	9,216
Maximum World Size	9,216	64	72

Source: Semianalysis

Key takeaway

Core Challenges of Scale-Up and the Inevitability of CPO

1. 目前主流的 Scale-Out CPO 雖解決了功耗與密度問題，但受限於跨機櫃網路的通訊協議延遲與頻寬瓶頸，仍無法支援跨機櫃的張量平行 (TP) 計算，算力孤島（以 Rack 為單位）的限制依然存在。
2. 在電連接方面，隨著單通道頻寬邁向 200G/lane 以上，銅線傳輸距離急遽縮短。即便使用主動電纜 (AEC) 勉強維持機櫃內 (Intra-rack, ~3m) 的連接，也會衍生出龐大的額外功耗，徹底失去提升叢集規模的物理可行性。

→Scale-up CPO 是運算叢集擴大的必然結果。

Technical Architecture and Implementation Paths for Scale-Up CPO

1. 與Scale-Out CPO相比，Scale-Up CPO對於ELS/FAU/OE的用量會成倍數增長，粗估會達到五倍。
2. 透過 Scale-Up CPO 將內部互連（如 NVLink）光學化，能成功打破實體機櫃的物理邊界。讓跨機櫃的 xPU 也能享有極大頻寬與極低延遲，徹底解放張量平行 (TP) 運算時的系統規模上限。

The Economics of Scale-Up CPO: Supply Chain Value Reconstruction

1. CPO OE中的E/PIC pair 會引入新的先進封裝及測試需求，並且在OE的組裝段，會引入Fab及OSAT加入競爭
2. Pluggable CM部分短期內不會改變的可能只有ELSFP的部分，ELSFP作為CPO system的消耗品未來的用量可期

Key takeaway

Technical and Commercial Risk Assessment of Scale-Up CPO

1.針對 GC 耦合的 FAU 是目前 CPO OE 的製造瓶頸之一。考量到 CPO 在後續封裝製程中極易發生光纖斷裂，具備可替換特性的 D-FAU (Detachable FAU) 將是未來拉升模組製造良率、降低沉沒成本的關鍵零組件。

2. 目前 CPO OE 的組裝高度仰賴耗時的主動對準 (Active Alignment, AA)。未來的封裝設計趨勢，將透過導入微光學元件來放寬 FAU 的物理製作公差 (Relax Tolerance)，並輔以 AA 進行最佳化耦合，以確保光學損耗 (Insertion Loss) 降至最低。

Investment Opportunities and Business Outlook

1.不管使用何種調變技術或 xPU 整合方式 (Shoreline Attach / Optical Interposer)，ELS、GC 與 FAU 皆已成為主流 Scale-Up CPO 解決方案的標準配備，為供應鏈帶來明確的投資價值。

2.對整體系統而言，引入 OCS (光電路交換) 來取代傳統電子交換機，可進一步突破傳輸延遲與能耗的瓶頸。有鑑於 Google TPU 架構結合 OCS 成功支撐了 Gemini 模型的強大算力，未來 OCS 在 AI 資料中心的大量商用化普及高度可期。

Thank You

for your time and attention.

回填問卷者
會後我們將會提供
本期QA精華內容

溫馨提醒檔案連結即將關閉
請把握時間下載

Please scan the QR code and fill out the survey.